

CHIRAG LAKHANPAL

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SUMMARY

Data Engineer with 5+ years of experience designing and implementing scalable data architectures and machine learning pipelines. Proven ability to build robust data solutions on Azure and Databricks, optimizing performance and delivering business value across healthcare, telecommunications, and regulatory compliance sectors. Expertise in PySpark, SQL, and streaming technologies like Apache Kafka.

PROFESSIONAL EXPERIENCE

Data Manager (Data Engineer)

Enhesa

August 2024–Present

Washington, D.C.

- Designed and implemented a Medallion architecture in Microsoft Fabric, leveraging Delta Lake to create Bronze, Silver, and Gold layers for 12+ regulatory data sources, ensuring data quality and consistency.
- Developed real-time customer segmentation pipelines using Apache Kafka and data integrated from Dynamics 365 CRM, processing **50,000 records/day** with **99.5% uptime**, enabling personalized marketing campaigns.
- Created a unified customer profile (UCP) by integrating data from Dynamics 365, ZoomInfo, and HubSpot into a Databricks-integrated Fabric data lake. Applied k-means clustering in Python to segment **over 2 million accounts**, contributing to a **12-15% improvement** in targeted upsell opportunities.
- Automated Total Addressable Market (TAM) calculations using PySpark in Azure Data Factory, processing **10 GB/week** of market data with a focus on meeting a **99.9% data availability SLA**.

Data Engineer

Patient-Centered Outcomes Research Institute (PCORI)

June 2023–April 2024

Washington, D.C.

- Migrated **17 TB** of historical healthcare data from Oracle Exadata to Azure Synapse. Optimized PySpark jobs on Databricks, resulting in a **30-35% reduction** in nightly batch processing time.
- Developed a BERT-based NLP pipeline on Databricks for clinical trial document parsing, achieving **80-85% accuracy** in metadata extraction.
- Built an XGBoost model** to predict grant application outcomes (\$2M threshold), utilizing **MLflow** for experiment tracking and PySpark for feature engineering.

Data Engineer

Tata Consultancy Services

March 2020–July 2022

Maharashtra, India

- Developed complex SQL stored procedures and materialized views for processing telecom data, incorporating window functions and dynamic SQL. This contributed to identifying at-risk customers, leading to targeted retention efforts and a **5-7% reduction** in churn. Processed **5 million daily** Home Location Register (HLR) records.
- Optimized 5G network provisioning by redesigning SQL queries with partitioned views and indexed temporary tables, achieving a **30-35% improvement** in data processing speed and reducing data retrieval latency to **under 3 seconds**.
- Automated data reconciliation for **23+ Apache Airflow DAGs** by implementing SQL-driven data quality checks using advanced joins and Common Table Expressions (CTEs). Reduced monthly ETL error rates from **8% to 4%**.

Junior Data Engineer

Tata Consultancy Services

March 2019–March 2020

Maharashtra, India

- Built and optimized complex SQL queries using window functions, CTEs, joins, and subqueries to process **8 TB** of Call Detail Record (CDR) data. Reduced data ingestion time by **60%**.
- Developed automated SQL stored procedures and views for customer churn analysis, reducing manual data processing effort by **approximately 15 hours/month** while maintaining **99.9% data accuracy**.

EDUCATION

Master of Science in Data Science The George Washington University	May 2024 Washington, D.C.
• GPA: 3.97/4.00 • Honors: Global Leaders Fellowship, Graduate Teaching Assistantship	
Bachelor of Engineering in Electronics and Communication Engineering Gujarat Technical University	June 2018 Gujarat, India
• GPA: 8.97/10.00	

PUBLICATIONS

Harish, S., Lakhanpal, C., & Jafari, A. H. (2024). Leveraging graph-based learning for credit card fraud detection. *Neural Computing and Applications*. doi.org/10.1007/s00521-024-10397-7

TECHNICAL SKILLS

Cloud Platforms	Azure (Data Factory, Fabric, Synapse, Databricks), AWS (Redshift)
Data Architecture	Medallion Architecture, Delta Lake, Kimball Dimensional Modeling
Big Data Technologies	Apache Spark (PySpark, Spark SQL), Apache Kafka, Apache Hive, ADLS Gen2
Databases	SQL Server, Oracle, (Add others if you're proficient - e.g., PostgreSQL, MySQL)
Programming Languages	Python, SQL, Scala, Java, R
Machine Learning/BI	scikit-learn, TensorFlow, Power BI, Tableau, Neo4j (*add if proficient: mlflow, DVC*)
Certifications	Azure Data Engineer Associate (DP-203)

TECHNICAL PROJECTS

Fraud Detection Graph System	GitHub — Publication
• Designed a graph neural network using PyTorch Geometric and DGL to detect credit card fraud, achieving an F1-score of 0.78. • Published in <i>Neural Computing and Applications</i> (Impact Factor: 5.1, from 2023).	
Healthcare NLP Pipeline	Azure — Python
• Built an end-to-end document processing pipeline using spaCy and Azure Functions to extract over 50 entity types from PDF documents.	
Real-Time Telecom Analytics	Spark — Kafka
• Created a streaming data pipeline to analyze millions of CDR events daily using Apache Spark Structured Streaming and Kafka, enabling real-time insights.	

HONORS AND AWARDS

- **TCS Digital Fast Track Promotion** (2020): Selected for accelerated promotion to TCS Digital, a specialized division focused on Fortune 500 clients, based on exceptional performance.
- **Humana-Mays Healthcare Analytics Competition**: Part of a team that achieved a top **10% ranking (53/750)** with a medication adherence prediction model (AUC: 0.81).
- **TCS Kudos Award** (2019): Recognized for designing an innovative ETL framework during training.
- Received 5+ **On-the-Spot Awards** for delivering critical data solutions that improved service level agreement (SLA) adherence.